

CHENGRU (ERIC) ZOU

400 Dowman Dr, Atlanta, GA 30322 • +1 (404) 954-0423

eric.zou@emory.edu • GitHub: @EricZou007 • Personal Web: @EZ

EDUCATION

BS in Applied Mathematics and Statistics

Emory University, Emory College of Arts and Sciences

Aug 2022 – May 2026
Atlanta, GA

- Cumulative GPA: 3.96/4.0; Honor: Dean’s List
- Expected: Summa Cum Laude (May 2026)

SKILLS AND INTERESTS

Research Interests: Scientific Machine Learning, Inverse Problems, Numerical Optimization.

Related Coursework: Iterative Methods (Grad), Numerical Linear Algebra (Grad), Complex Analysis (Grad), Real Analysis, Numerical PDEs, Generative Model.

Skills: Python, PyTorch, MATLAB, R, SQL, Git, Java, C, HTML.

Toolkit: Docker, SSH, AWS.

PUBLICATIONS

- Chengru Zou**, Elle Buser, Zichao Wendy Di, Yuanzhe Xi. *A Joint Variational Framework for Multimodal X-ray Ptychography and Fluorescence Reconstruction*. arXiv:2511.02153 [math.NA], 2025. arxiv.org/abs/2511.02153
- Erise He, Zhijiang Ye, **Chengru Zou**. “Quantitative Hybrid Structure Analysis of Equipment Wear under Dynamic Pressure based on Digital Intelligent Algorithm.” *Proceedings of the IEEE International Conference on Data Science and Intelligent Systems (ICDSIS)*, 2025, pp. 1–7. DOI: [link].

RESEARCH EXPERIENCE

Research Assistant: Joint X-ray Reconstruction for Multimodal Imaging

Emory Math Department & Argonne National Laboratory

Aug 2024 – Present
Atlanta, GA

- Designed a physics-informed joint reconstruction framework for X-ray ptychography and fluorescence modalities under the guidance of Dr. Wendy Di and Dr. Yuanzhe Xi.
- Analyzed optimization behavior via loss surface geometry to characterize the highly nonconvex objectives in single and multimodal reconstructions.
- Implemented (L-BFGS, Truncated Newton–Krylov, NLTGCR) numerical solvers with robust line search and stopping criteria in MATLAB for non-convex optimization.
- Demonstrated up to 30% lower MSE and 10% lower loss versus single-modality baselines on standard benchmarks; validated across overlap settings and noise levels.
- Published the manuscript to arXiv as the first author and submitted to *SIAM Journal on Imaging Sciences*.

Research Assistant: Multilevel Block Adapter

MIT-IBM Watson AI Lab

July 2024 – Present
Atlanta, GA

- Developed *Multilevel Block Adapter (MBA)* for parameter-efficient fine-tuning based on the multilevel method and spectral clustering property of LoRA weight matrix under the guidance of Dr. Jie Chen.
- Performed algorithmic analysis and experiments on V100 GPU to derive NLU/NLG-specific learning-rate schedules for *MBA*.
- Implemented reproducible cloud experiments pipeline through SSH and Docker; published on GitHub with saved checkpoints and scripts.
- Introduced a variational parameterization of the *MBA* weight matrix, reducing adapter parameters on RoBERTa-base and GPT-2 from 300k to 120k while achieving 2% higher benchmark scores on GLUE and E2E.

PROJECTS

AI Chatbot Agent Development

June 2024 – Aug 2024

Emory Center for AI Learning, Emory University

Atlanta, GA

- Built a graphical RAG LLM chatbot agent for Emory student course selection using Neo4j graphical database and LangChain prompt engineering.
- Achieved up to 20% improvement in query retrieval accuracy over the vector RAG model, benefiting 100+ of Emory students for course selection.
- Published the report and gave a presentation in the Emory data science department symposium.

Atlanta Affordable Housing

Sep 2023 – Jan 2024

City of Decatur: Affordable Housing Commission

Atlanta, GA

- Collaborated with group leader and members to address over 1000 housing complaints in partnership with the City of Decatur, GA.
- Utilized Geocoding and GIS to model racial and housing rate distributions in the great Atlanta Area.
- Presented findings using Tableau, aiding over 100 local residents in making informed housing decisions.

AWARDS AND GRANTS

Mathematical Contest in Modeling (MCM), Meritorious Award

May 2025

Modeled stair erosion to estimate construction age under the supervision of Dr. Alessandro Veneziani; ranked in the top 5% among 20,000+ participating teams; published the award-winning paper at the *IEEE* conference.

Emory Pathway Global Internship Grant

May 2024

Awarded \$4,000 for outstanding summer internship performance.

American Statistical Association (ASA) DataFest Competition, Finalist

April 2024

Analyzed real-life raw data from CourseKata Company; provided consulting advice that increased study efficiency for over 1,000 students.

Emory Phi Sigma, Member

May 2023

Nationalwide freshman honor society; invitation based on scholastic achievements during freshman year.

PROFESSIONAL SOCIETY MEMBERSHIP

Atlanta Preconditioning 2024

Student Volunteer

Helped organize the academic conference in the Atlanta Area.

Atlanta Mathematics of Data Science Bootcamp

Attendee

Studied Lipschitz-regularized χ^2 -GANs combining χ^2 -GAN and WGAN.

Atlanta Science Festival – Exploration Exposition

Student Volunteer

Promoted mathematical knowledge to the public on behalf of the Emory Mathematics Department.